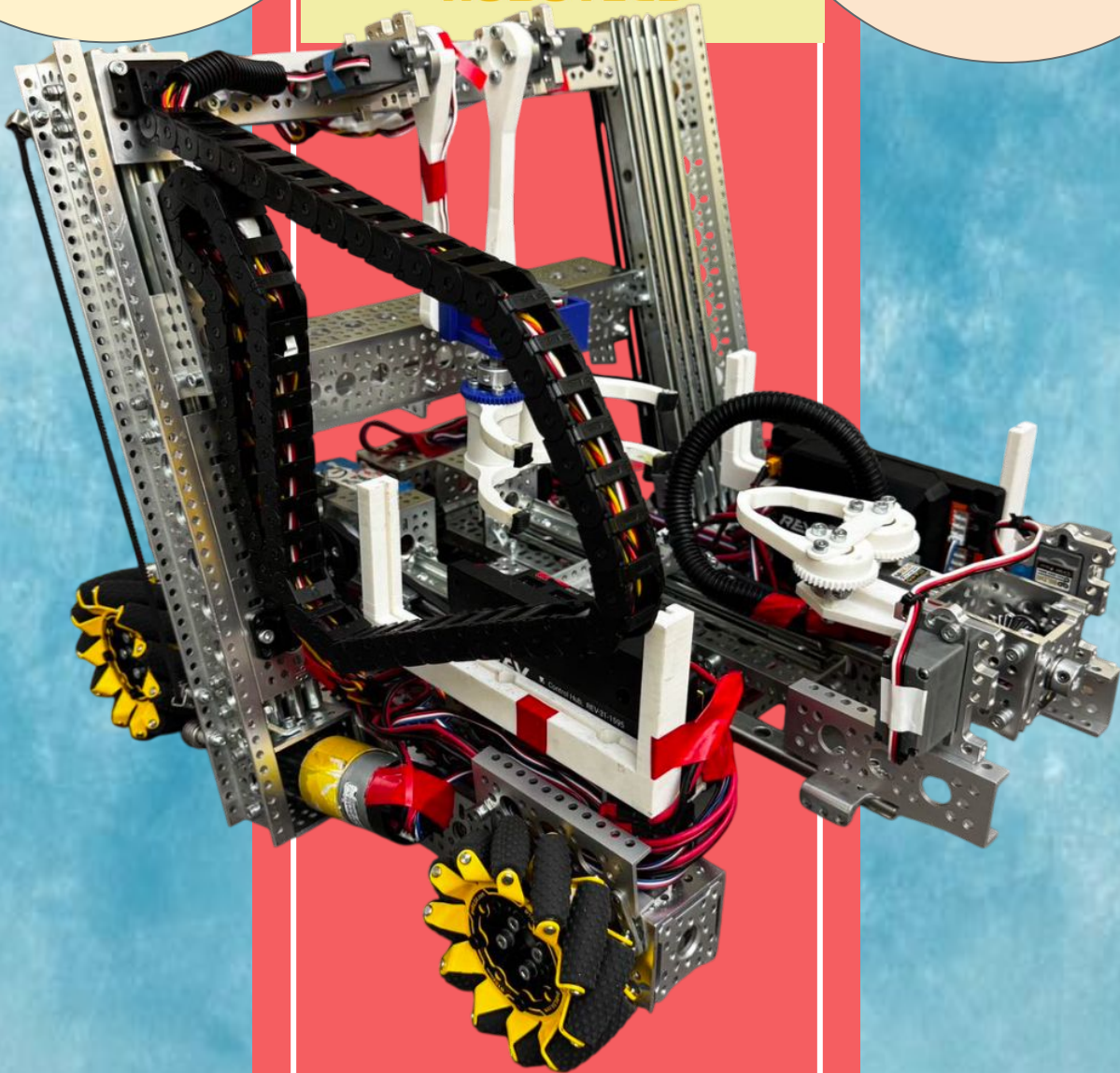


FTC Team  
23365



**ALPHA  
ROBOTICS**

Engineering  
Portfolio



# Catalog

1. team spirit
2. Specs about our team
3. Designing Process
- 4-7. Robot Building Process
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## Our team spirit

Our FTC robotics team embodies the spirit of collaboration, innovation, and determination. Every challenge we face becomes an opportunity to learn and grow, as each team member contributes their unique skills and perspectives. From brainstorming solutions to fine-tuning our robot, we foster an environment where creativity and teamwork thrive. With every competition, we not only aim to perform at our best but also to inspire and support each other, building lasting friendships and a shared commitment to excellence. The true strength of our team lies in our collective drive to overcome obstacles, adapt to new ideas, and celebrate both our successes and the lessons we learn along the way.

## How our team theme is supportive of the *FIRST* Core Values.

**Discovery:** Our team has the Excellent qualities of active experimentation and not being afraid of facing difficulties. We always explore new ideas and try to improve our robot. We watch tons of robotics videos from other teams online. If we find some good design or scoring concept, our team will discuss together and try to learn from their design

**Innovation:** Our ultimate goal is to create new and innovative designs. We follow the design process to create innovative designs such as our claws.

**Impact:** Doing robotics brings us many advantages such as enhancing our problem solving and critical thinking skills and fostering teamwork and collaboration skills. These skills are essential for us to make a big success in any field which will probably help the world.

**Inclusion:** In our team, being kind and respecting each other was extremely important. For example, in a discussion, everyone will Listen patiently to every sentence although they have opposite opinions. During the competition, we will not deliberately belittle or praise the design of any team. We will think rationally and learn from it.

**Teamwork:** At the beginning of the season, everyone was not good at teamwork. They did not cooperate and assist each other. Instead, everyone did what they liked. However, after several months of working together, all the team members have developed a very tacit understanding.

**Fun:** The charm that robotics brings to us is unparalleled. We feel the fun of constructing and competing.

# Specs about our team

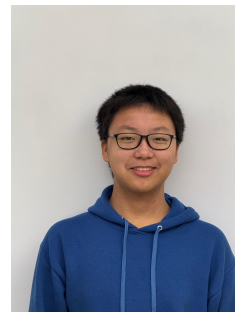
## Our Team ( 2024-2025 )

Alpha Robotics (team 23365) is a second-year robotics team. (started in 2023) This year our team has 3 eighth graders(2 are new), and 4 high school students(1 is new), who are all interested in science and technology. At Team 23365, we firmly believe that the foundation of any successful robotics project lies in the perseverance of our team and the collaborative spirit that drives us to innovate and improve. The spirit of our team is well reflected in the process of building the robot and our design.

## Team Member Categories

|            |                                                    |
|------------|----------------------------------------------------|
| Robert L   | Outreach and Builder                               |
| Jason Z    | Driver and Builder                                 |
| Maximus X  | Driver and Builder                                 |
| Steve L    | Human Player and Builder                           |
| Jason S    | Team Captain<br>Programming Captain<br>and Builder |
| Andy S     | Programmer and Builder                             |
| John Chen  | CAD and building<br>Captain                        |
| Zhaolong S | Coach and Mentor                                   |

Jason Z



Maximus



Steve



Andy



Coach Zhaolong



Zonglin L



Jason S



John



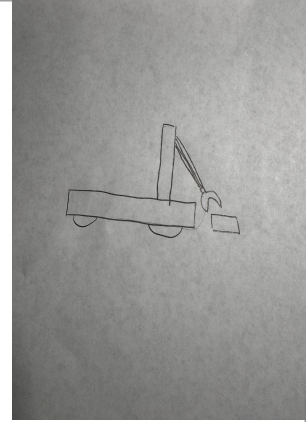
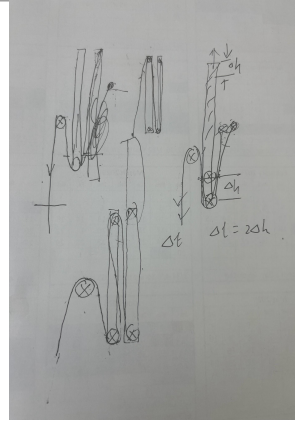


## Goals of our team this year

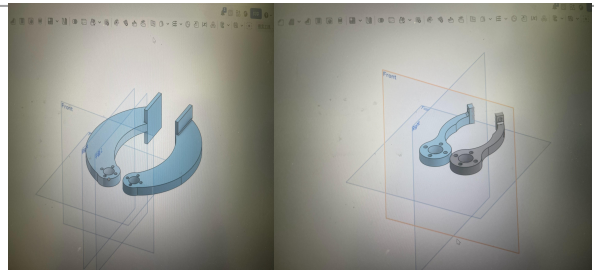
- 1 Go to state championship this year
- 2 help new members learn engineering and programming skills
- 3 become top 100 in the world

## Designing Process

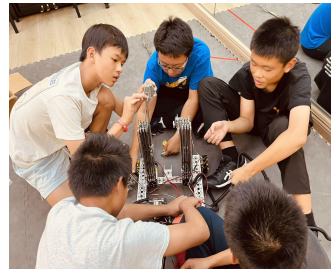
**Identifying and Brainstorming:** The coach immediately assigned our jobs after the season started. . We gathered together to discuss scenarios, tactics, and possible problems and difficulties with the robot immediately after the topic was given. Everyone is active in giving their opinion. Draw designs on paper or white board.



**Prototype and Design :** We start prototyping after brainstorming. We used onshape to design parts on our robot such as intake and outtake claw and test the functionality of the design.



**Assemble:** after creating prototypes, we assemble the components together to see if it functions



**Testing:** After constructing our robot, we need to test our robot to see how it works in order to find out the issues and the solutions



**Improve (interpret )** After we found out issues and weaknesses of our robot. We need to create a better design or build a new robot in order to improve our design.

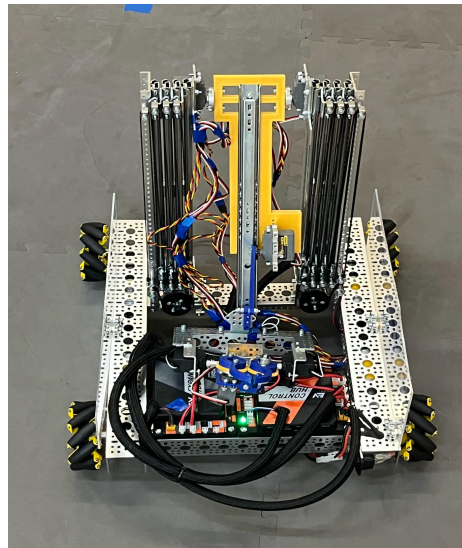
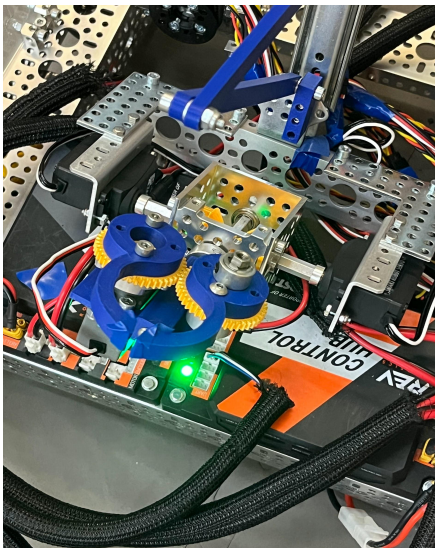




# Robot Building Process 1

**First Generation:** We used a linear slide system, with a vertical slide that can bring things up high and a pivoting slide that can reach things in the submersibles. We designed a linkage brick that connects our reaching slide and claw to the main vertical slide pair. The linkage brick is connected to the vertical slide pair by two servos allowing it to pivot up and down. When the linkage brick wasn't ready yet, our coach helped us create a non custom one using Gobilda parts. (We finished building our first robot before meet 0.)

Picture of the claw attached to differential gear on the linkage



Linkage Brick ( middle 9 holes attached to reaching slides, 4 side holes attached to servos.)



# Robot Building Process 2

## Improving Process ( First Robot )

The first robot did not do well in meet 0. The pivoting slide wasn't working because the servo that actuates it doesn't have enough power. The center of mass was high and the linkage brick was heavy. We researched on robot designs and found a new solution to the game, so we were determined to build a new robot. However, before we start building our second robot. We need to find out the issues and solutions.

### Issues:

The arm was too heavy so the motor of the linear slides could not lift it up. (weight)

The claw could not grip well (claw design)

The connection lines that connect the control hub and the electronic component were too messy and were easily damaged and disconnected. (cable management)

Couldn't hang(endgame)

Our team's number plate does not meet tournament requirements. (number plate)

Overall, the design failed

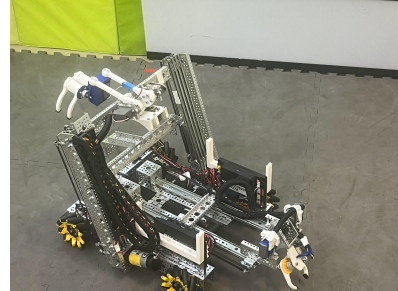
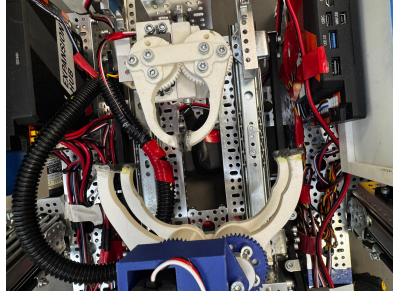
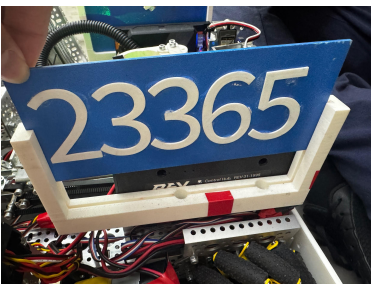
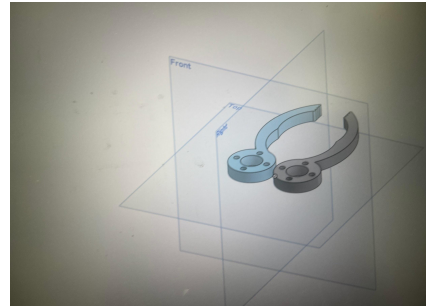
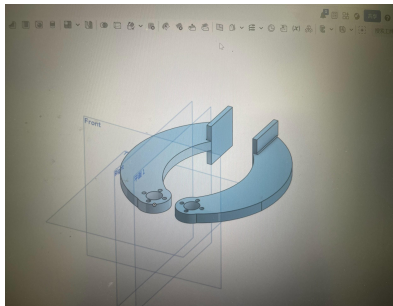
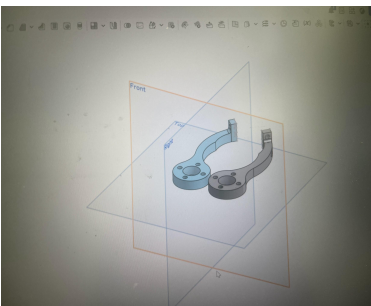
### Solutions:

We thought that we needed to build a robot with two sets of linear slide pairs and better cable management. One slide can go horizontal to grab samples. The other slides can go vertically and place the sample into the basket. (higher efficiency and less workload for each part)

Designed new claws and added rubber at the end of each claw for grip

Redesigned number plate and holder for it

### Pictures of our team changing and improving our robot





# Robot Building Process 3

## Our Second Robot (Current robot):

We started building our second robot after meet 0. We thought that scoring samples would be easier and quicker, so our second robot mainly scores samples. We spent nearly 14 hours every week on robotics. After three weeks of hard work, our team finally finished building our second robot. Our second robot uses four 312 mph motors to power our drivetrain and mecanum wheels(can strafe). For our linear slides, we use two 435 mph motors for the vertical linear slide pair and one 1080 mph for the horizontal pair(show in image below). A U-channel is connected to the vertical linear slide pair with a hook(bottom of U-channel) and a specimen claw(top of u channel) that clamps onto specimens. The robot performed well in meetings 2 and 3. Our team created the highest match score with Bread Pandas in our league which was 277 points, and we can score an average of 12 samples during teleop in our practice sessions and scored the highest of 12 samples during meet 3. We were satisfied with the result. However, we know that this isn't the best that we can do. We still need to improve our robot.

### Strengths and Weaknesses

😊 Using two stages linear slides can score the point really fast (efficiency)

😊 claw can precisely pick up samples that you aim for (precision)

Disadvantages :

😞 Our robot and linear slides were too heavy so the speed was very low compared to other teams. (Weight and speed)

😞 horizontal claw limited. Cannot reach far enough into the submersible, limited selection of samples

😞 center of mass not low enough, causes little wobbling during moving and reaching slide upwards

😞 robot is too bulky, making moving in small places difficult(size)

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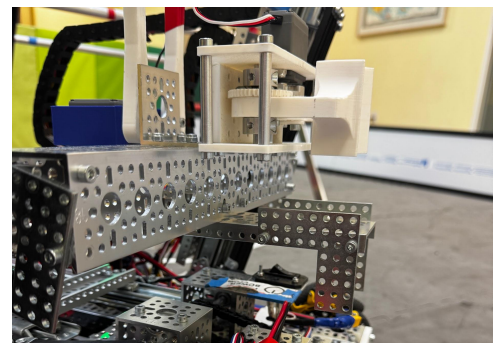
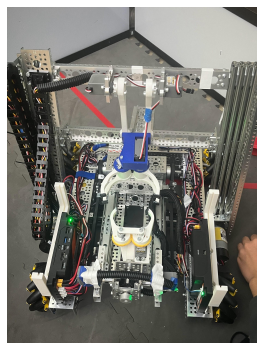
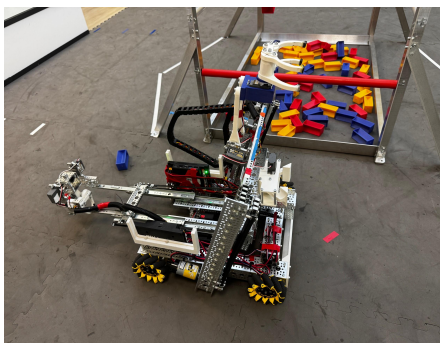
### Robot's features on appearances

**Structure:** Our robot is an axisymmetric structure, with the same structure on both sides, including the battery and control terminal. In the middle is the scoring catcher and telescopic part.

**Neatness and organization :** Depending on where the wires go, use zip ties and tape to tie them together, which prevents wire damage and disconnections, also makes robot look more neat

**Atmospheric texture :** The complex construction of the robot's interior has a beautiful texture and atmospheric demeanor.

**Color matching:** In terms of appearance, the main body of our robot is made of silver-white metal. For delicate parts such as gears and clips that need to be 3D printed, we use only white color. Which looks powerful and domineering.



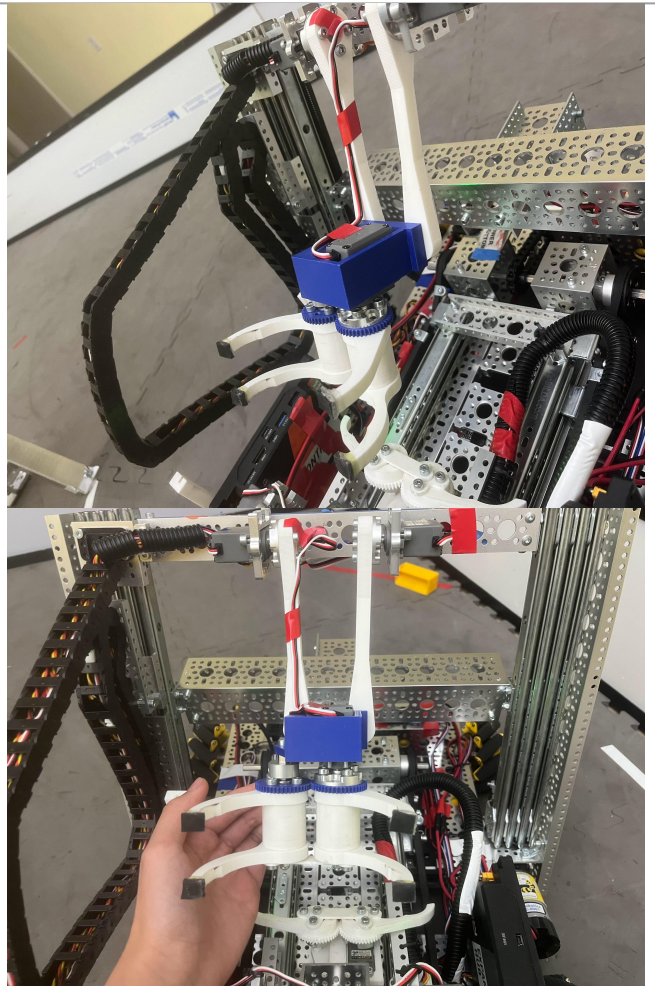


# Robot Building Process 4

## The most Innovative Design

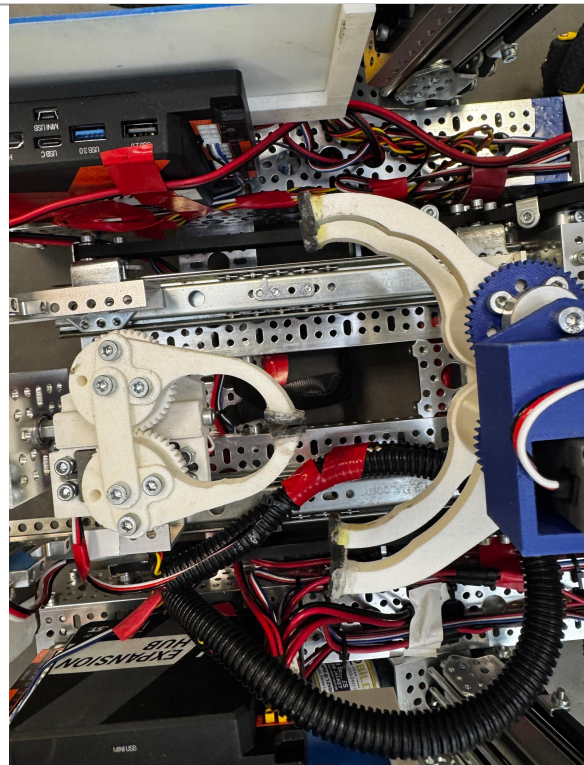
### The upper claw

The upper claw, which we use to drop samples, is attached to the inside of the vertical linear slide pair by a two servos and two arms. People usually approach this claw using only two “fingers”, but we use 4 as shown in the picture. This is similar to our hands since we have multiple fingers to grab things for more stability and control. So even if the upper two “fingers” don’t clamp onto the sample, the lower two “fingers” can prevent the sample from dropping out of the claw and still hold on to it. Compared to other robots with only two “fingers” as their outtake claw, our claw can grab on things more easily.



### The lower claw

Most teams that use passive intake approach the lower claw using super small claws. These small claws are good at grabbing from places where 2+ samples are consecutively touching each other’s faces, but they require higher precision in placing the claw which is hard for drivers. Our claw doesn’t need the driver to be super precise because it is larger, and this makes our sample pickup time quicker and our drivers can have lesser errors which creates confidence. They work efficiently in almost all cases.

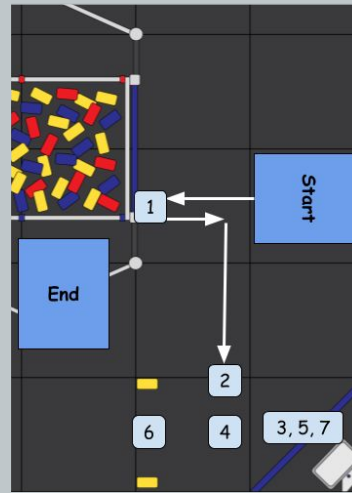


# Programming-Autonomous

## Autonomous Overview

Our goal for autonomous this year is to reliably score 3 samples into the high basket. In order to achieve reliability our robot needs a way to accurately determine its position at all times. This lead to our decision to use 3 wheel odometry to achieve accurate positioning.

## Autonomous Path

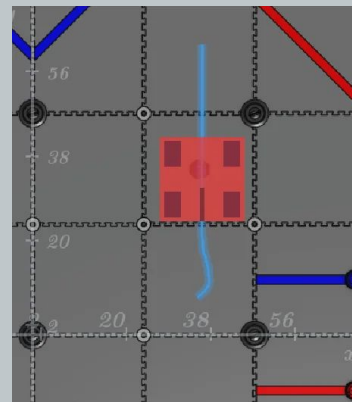


1. Place specimen at 1
2. Go to 2 and pick up a sample
3. Deposit sample at 3
4. Go to 4 and pick up a sample
5. Deposit sample at 5
6. Go to 6 and pick up a sample
7. Deposit sample at 7
8. Go to End position

## Odometry

Odometry is a technique that uses the data of the object's own motion to calculate the current coordinates of an object. For example, the 3-wheel odometry we use calculates the difference between the robot's current X and Y coordinates and the starting position by counting the number of turns the wheels make.

## Roadrunner



Roadrunner is an advanced motion profiling library developed for FTC which receives data from our odometry dead wheels to analyze the position and construct an optimal path to our predefined coordinates. This allows us to quickly and consistently move along our paths in autonomous.

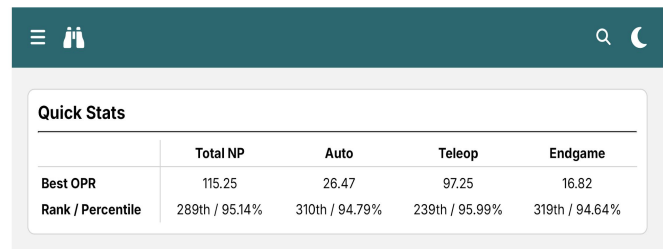
# Match Performance

In autonomous, we score 1 specimen and 3 samples, total of 34 points

In teleop, we grab samples from the submersible and put them to the higher basket. We average around 12 in our latest practice matches. We score total of 15 samples and 1 specimen(3 samples and 1 specimen carried on from auton). 130 points in teleop.

Endgame: We do a level 2 ascent scoring us an extra 15 points

Stats of our team from ftc scout:



The screenshot shows the FTC Scout app interface. At the top is a dark teal header with a menu icon, a team icon, a search icon, and a moon icon. Below the header is a 'Quick Stats' section with a table. The table has five columns: an empty header cell, 'Total NP', 'Auto', 'Teleop', and 'Endgame'. The first data row shows 'Best OPR' with values 115.25, 26.47, 97.25, and 16.82. The second data row shows 'Rank / Percentile' with values 289th / 95.14%, 310th / 94.79%, 239th / 95.99%, and 319th / 94.64%.

|                   | Total NP       | Auto           | Teleop         | Endgame        |
|-------------------|----------------|----------------|----------------|----------------|
| Best OPR          | 115.25         | 26.47          | 97.25          | 16.82          |
| Rank / Percentile | 289th / 95.14% | 310th / 94.79% | 239th / 95.99% | 319th / 94.64% |

Highlight of our team's performance :

We won all 5 matches on meet 3 .

We are now second place in our league and 293 place in the world(1/23/25)

We create the highest score with team 21980 Bread Pandas during meet 2(277 points)

We also created the highest score in meet 3 with team 21990(209 points)

Highest solo score in meet 3: 171 points

Highest solo score in practice: 195 points

Average solo score in practice: 179 points



# Outreach

Orange County makers faire: We attend this event and show visitors our robot and team information. Through this event, our team members saw lots of cool things from other people and learned from colleges, hobbyist, and local businesses.



# Impressions and Conclusions

## Our Impressions

All of us learned and improved a lot during this year's work and competition. Our team's core strength lies in our unwavering perseverance and relentless drive for innovation. We face each obstacle with a determination to push through, knowing that every setback is just another opportunity to improve. Whether it's overcoming technical challenges or refining our designs, we approach each task with creative thinking and an open mind, always eager to try new solutions. This spirit of continuous improvement fuels our progress, and we take pride in turning ideas into reality through hard work and resilience. Innovation isn't just about building a better robot; it's about developing a mindset that never settles, always strives for the next breakthrough, and believes that with enough effort, any challenge can be conquered. We also surely argue that the power of success is rooted in a deep faith that with dedication, effort, and teamwork, we can overcome any challenge.

## Future Plans

We will continue to participate in the FTC competition and encourage other teenagers to become a member of the community. We want to go to worlds next year and make our team consistently go to worlds all the years after, inspiring hundreds of teenagers

## Conclusions:

In conclusion, our robotics team's engineering portfolio reflects not only the technical expertise and innovation we bring to every project, but also our commitment to continuous learning and improvement. Through collaborative problem-solving, creative design, and persistent iteration, we've transformed challenges into opportunities, pushing the boundaries of what we can achieve. This portfolio stands as a testament to the hard work, resilience, and teamwork that have driven our success, and serves as a foundation for future growth and exploration. As we continue to refine our skills and adapt to new technologies, we remain focused on the goal of making a positive impact in the world of robotics and beyond.